8 Learning principles CountMeInTT uses

1. **Retrieval practice ("testing effect")**

- **What/Where it shows up:** Students who are now learning can re-read the answers. The challenge comes in when the speed improvement factor is encouraged. Every grid forces students to pull multiplication facts from memory (no longer re-reading or copying), in order to improve their time.
- **Why it works:** Active retrieval strengthens long-term retention more than passive study.

2. **Spaced / distributed practice + repetition**

- **What/Where it shows up:** Students return across days/weeks, log multiple attempts, and compare improvement over time.
- **Why it works:** Spacing practice beats cramming for durable learning.

3. **Deliberate practice with a clear feedback loop**

- **What/Where it shows up:** "Time + accuracy" gives a measurable result; students set targets, retry, and refine weak areas.
- **Why it works:** Repeated practice with feedback is a core driver of skill improvement.

4. **Fluency & automaticity (speed after accuracy → frees working memory)**

- **What/Where it shows up:** Timed repetition builds faster recall so students can handle harder, multi-step math with less mental strain.
- **Why it works:** Automatic basic facts free working memory for reasoning and problem-solving.

5. **Immediate feedback + clear error signals**

- **What/Where it shows up:** Wrong entries are caught right away; students must correct them to complete the grid. Red squares on the grid for incorrect answers and green squares for correct answers. Students must press the enter button to confirm their input.

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- **Why it works:** Timely, usable feedback is one of the highest-impact learning supports.

6. **Goal-setting + self-monitoring (personal best tracking)**

- **What/Where it shows up:** “Beat your own time,” track attempts, track progress, set a target time for the next attempt.
- **Why it works:** Clear goals + visible progress increases persistence, confidence, and performance.

7. **Gamification + goal orientation (levels, leaderboards, challenge structure)**

- **What/Where it shows up:** Levels (5×5 → 5×12 → 12×12 → 15×15), podiums, team challenges, prizes, hall of fame.
- **Why it works:** Gamification tends to increase participation and engagement—especially when tied to progress and competence.

8. **Formative assessment / diagnostic value (“micro checks”)**

- **What/Where it shows up:** Teachers/tutors/parents can spot exactly where students hesitate, revealing weak tables/facts without a formal test.
- **Why it works:** Low-stakes ongoing checks guide targeted practice and intervention efficiently.

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